Car Rental System Case Study - conditional statement case study

Task 1: Define Rental Rates Based on Car Type

Objective: Set different rental rates for car types (Economy, Standard, Luxury) using conditional statements.

1. Prompt the user to enter a car type (Economy, Standard, or Luxury).
2. assign a daily rental rate:

- - Economy: $50/day
- - Standard: $75/day
- - Luxury: $100/day

3. If an invalid car type is entered, display an error message and exit.

Expected Output: The daily rental rate based on the car type

Task 2: Calculate Rental Duration Fee

Objective: Calculate the base rental fee by multiplying the daily rate by the number of rental days.

1. Prompt the user to enter the number of rental days.
2. Validate the input as a positive integer; otherwise, display an error and exit.
3. Calculate the base_fee as the product of the daily rental rate and rental days.

Expected Output: The base_fee based on the rental days and car type

Task 3: Apply Membership Discount

Objective: Adjust the base_fee by applying a membership discount if applicable.

Steps:

1. Prompt the user to enter their membership type (Premium, Standard, or None).
2. Use conditional statements to determine the discount rate:

- Premium: 15% discount
- Standard: 10% discount
- None: No discount

3. If the membership type is invalid, display an error and exit.
4. Calculate the discount amount and subtract it from the base_fee.

Expected Output: The adjusted fee after the membership discount.

Task 4: Add Optional Services

Objective: Calculate additional charges for optional services (Insurance and GPS).

Steps:

1. Prompt the user to indicate if they want to add Insurance (yes or no).

- - If 'yes', calculate Insurance fee at $20 per day.

2. Prompt the user to indicate if they want to add GPS (yes or no).

- - If 'yes', calculate GPS fee at $5 per day.

3. Use conditional statements to add the appropriate fees based on user choices.

Expected Output: The additional services fee based on selected options.

Task 5: Calculate Late Return Penalty

Objective: Add a penalty fee for any late returns.

Steps:

1. Prompt the user to enter the number of days the car was returned late.
2. If the car was returned late (greater than 0 days), calculate a penalty of $30 per late day.
3. Use conditional statements to apply this penalty only if late days are entered.

Expected Output: The total late return penalty

Task 6: Calculate and Display Total Rental Fee

Objective: Summarize all fees and display the total rental cost.

1. Sum up the base_fee, membership discount, additional services, and late penalty.
2. Display a detailed breakdown of the costs, including:

    - Car type and daily rate
    - Number of rental days
    - Membership discount
    - Additional services fee
    - Late return penalty
    - Total rental fee

3. Format the output for clear and readable presentation to the user.

Expected Output: A summary of all charges with a clear breakdown of the total rental fee.